MIL-I-23972C(YD)
21 May 1984
SUPERSEDING
MIL-I-23972B(YD)
10 February 1976

MILITARY SPECIFICATION

INSULATOR ASSEMBLIES, RACK, SECONDARY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical insulator assemblies consisting of four porcelain spool insulators, mounted on steel rack-back, for supporting secondary and service wiring, rated up to 600 volts potential, on poles and building structures.

1.2 Classification. Insulator assemblies shall be of the following types, as specified (see 6.2):

Type I - Medium-duty service.
Type II - Heavy-duty service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5970

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C29.1 - Electrical Power Insulators, Test Methods for.
C29.3 - Wet-Process Porcelain Insulators (Spool Type).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

C135.7 - NEMA Standards for Spool Type Secondary Racks.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The insulator assemblies shall consist of four porcelain spools mounted by means of steel legs and button-head pin to a steel rack-back as specified herein, and as shown in figure 1.

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3.2 First article. When specified (see 6.2), the contractor shall furnish sample insulators for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Codes and standards. The insulators shall conform to the requirements of NEMA C135.7 as applicable.

3.3.1 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the insulators, he proposes to furnish under this specification meets the requirements of NEMA C135.7 as applicable.

3.4 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.5 Type I insulator assembly. The type I insulator assembly shall be in accordance with NEMA C135.7, type MX-4S-8, except that a 1/2 inch button-head pin may be used provided the assembly is designed for a 1/2 inch button-head pin and conforms to all other requirements of NEMA C135.7, type MX-4S-8.

3.6 Type II insulator assembly. The type II insulator assembly shall be in accordance with NEMA C135.7, type HX-4S-8.

3.7 Surface protection. All ferrous metal parts shall be zinc-coated in accordance with ASTM A153, with a coat thickness capable of resisting the tests required in 4.5.1.

3.8 Insulator. Insulator shall be wet-process porcelain conforming to ANSI C29.3 spool type, as specified (see 6.2). Unless otherwise specified (see 6.2), color shall be brown glazed.

3.8.1 Low-frequency dry-flashover rating. Insulator shall be rated for a low-frequency, dry-flashover value of at least 20,000 volts when tested as specified in 4.5.2.

3.9 Insulator assembly strength. The overall strength requirements for type I and type II insulator assemblies shall be not less than the minimum shown in table II of NEMA C135.7.

3.10 Workmanship. The quality of workmanship shall be such as to produce insulator assemblies that are in accordance with the requirements of this specification and are so fabricated as to insure proper functioning of the unit.

3.10.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.10.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.10.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.10.4 Hot-upsetting. The hot-upsetting technique is where the standoff pin is the rivet and is secured by inserting it through the hole in the rack and hot-upsetting the protruding end.

3.10.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.1.2 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standards cited in 3.3.1.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on insulator assemblies when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the first article pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.3, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All insulators offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 4.0 percent defective.

4.4 Examination. Each sample selected shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements shall constitute cause for rejection.

4.5 Tests. Each sample selected in accordance with 4.3.2 shall be tested in accordance with 4.5.1, 4.5.2, and 4.5.3.

4.5.1 Galvanizing test. All ferrous metal parts of the sample shall be tested for uniformity of zinc-coating by the method specified in ASTM A153 and shall withstand two dips before reaching the end point. A properly calibrated magnetic or electromagnetic gage may be used to determine the zinc-coating thickness as required by 3.7.

4.5.2 Low-frequency dry-flashover insulator test. The insulators shall meet the low-frequency, dry-flashover values indicated in 3.8.1 when tested in accordance with the low-frequency withstand voltage tests specified in ANSI C29.1.

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4.5.3 Strength test. Insulator assemblies shall be tested to determine conformance to the strength requirements of 3.9.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified a first article pack shall be furnished for examination and test in accordance with MIL-E-17555 within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. In addition to the usual outside applications on pole and building exteriors, the insulator assemblies will be used within buildings, suspended horizontally or vertically.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title number and date of this specification.
b. Type of insulator assembly required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. Class of spool type insulator required (see 3.8).
e. When insulator is required to be other than brown glazed (see 3.8).
f. When a first article pack inspection is required and time frame for submission (see 4.6).
g. Level of preservation and level of packing required (see 5.1).

6.3 Information figure. Figure 1 shows a type of insulator assembly which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another insulator assembly which conforms to this specification.

6.4 First article. When a first article inspection is required (see 6.2), the item will be tested and should be a sample selected from the first insulator assembly production items or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article (see 3.2).

6.5 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the

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provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodian:

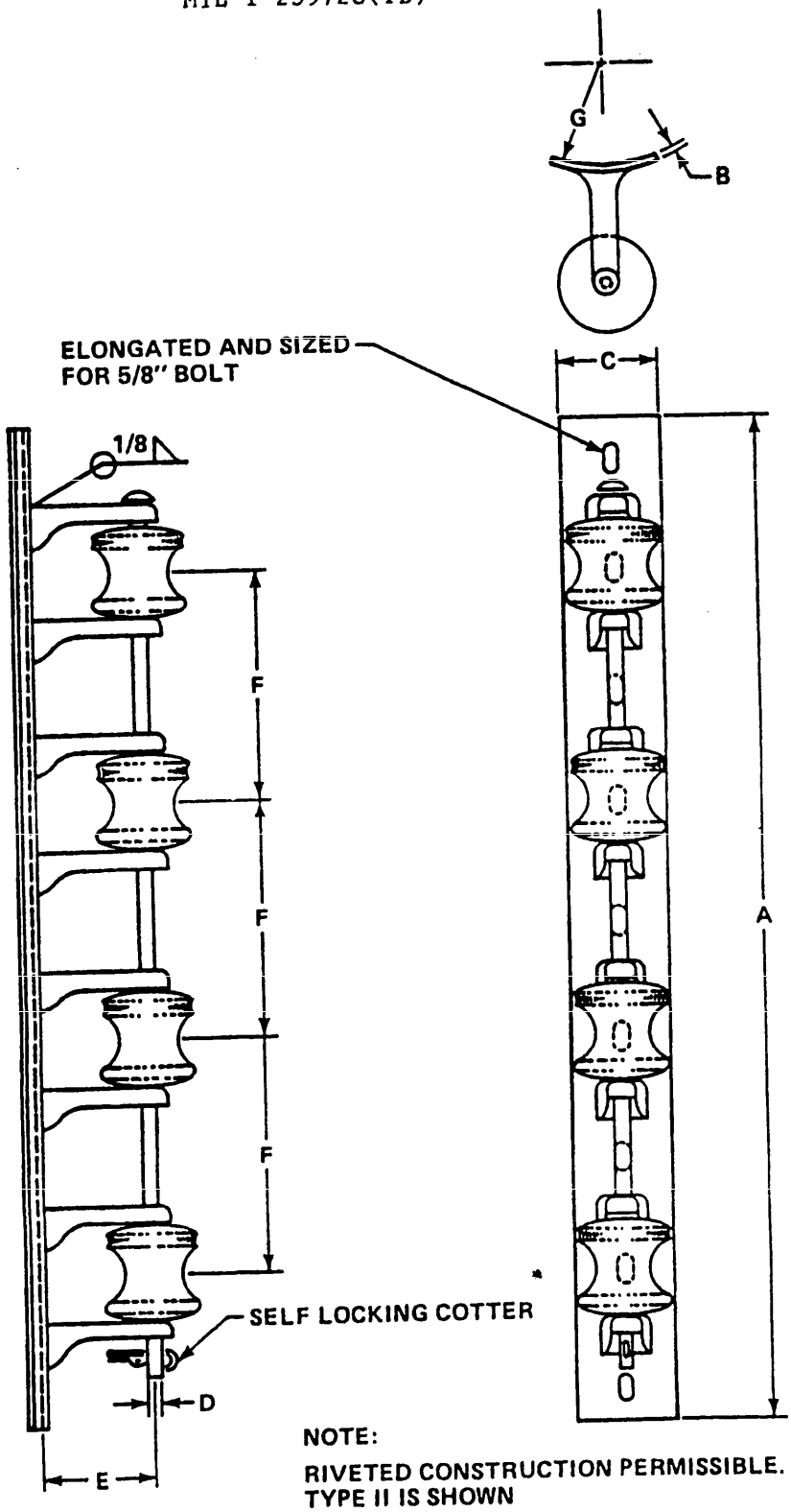
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	DIMENSIONS						
	A LENGTH OF RACK BACK	B THICKNESS	C WIDTH	D CLEVIS ROD DIMENSION	E DISTANCE FROM BACK TO ROD	F CENTER TO CENTER OF SPOOL	G RADIUS
TYPE I	34.0	0.1046	2.75	0.5	4.0	8.0	3.5
TYPE II	34.0	0.125	3.00	0.625	4.0	8.0	3.5

Figure 1.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23972C(YD)	INSULATOR ASSEMBLIES, RACK, SECONDARY

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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